Essential Summary

A Generalized Padeé Approximation Method for Solving Homoclinic and Heteroclinic Orbits of Strongly Nonlinear Autonomous Oscillators

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Chinese Physics B, Vol. 23, No. 12, 120501 (2014) DOI: 10.1088/1674-1056/23/12/120501

1 Background and Motivation

Classical Padeé approximation has been widely applied in nonlinear dynamics. Various researchers extended it by replacing polynomial components with exponential functions to better capture asymptotic homoclinic decay. However, these modifications were proposed case-by-case without a **unifying theoretical framework**.

This paper (the English version of the GPA method) proposes an intrinsic extension: numerator and denominator are generalized from polynomials to **series of any function type**, unifying all prior modifications. It further extends the scope to **both homoclinic and heteroclinic orbits** across three oscillator classes.

2 Core Innovation

2.1 Generalized Definition

Let $\widehat{H}_m = \{P : P(z) = \sum_{i=0}^{m} a_i g_i(z)^i\}$ extend H_m. The **GPA**$[L/M]_f$ is $P_L/Q_M \in G(L,M)$ satisfying $f(z) - P_L/Q_M = \mathcal{O}(z^{L+M+1})$. Classical Padeé ($g_i(z) = z^i$) and all quasi-Padeé variants become special cases.

2.2 Two Novel Hyperbolic-Function Approximants

For $\ddot{x} + g(x) = 0$ with saddle $H(h, 0)$:

Homoclinic ($x(\pm\infty) = h$):

$$\boxed{\mathrm{GPA}_{\mathrm{hom}}[L/M] = \frac{\sum_{i=0}^{L} \alpha_i \operatorname{sech}^i t}{1 + \sum_{i=1}^{M} \beta_i \operatorname{sech}^i t}}. \tag{1}$$

Heteroclinic ($x(\pm\infty) = h_{1,2}$):

$$\boxed{\mathrm{GPA}_{\mathrm{het}}[L/M] = \frac{\sum_{i=0}^{L} \alpha_i \tanh^i t}{1 + \sum_{i=1}^{M} \beta_i \tanh^i t}}. \tag{2}$$

Coefficients $\{\alpha_i, \beta_i\}$ are found by Taylor-expanding at $t = 0$ and matching with the power series $\{a_i\}$ from substituting $x(t) = \sum a_i t^i$ into $\ddot{x}+g(x) = 0$, with the initial amplitude from $\int_h^a g(x)dx = 0$. The hyperbolic basis functions automatically satisfy the asymptotic boundary conditions.

3 Key Results

3.1 Cubic–Quintic Duffing Equation

$$\ddot{x} + c_1 x + c_3 x^3 + c_5 x^5 = 0, \qquad c_1 = 2,\ c_3 = -10,\ c_5 = 9. \tag{3}$$

Five fixed points: three centers, two saddles. Two homoclinic orbits and one heteroclinic pair coexist. GPA[4/4] and GPA[7/7] capture all orbits.

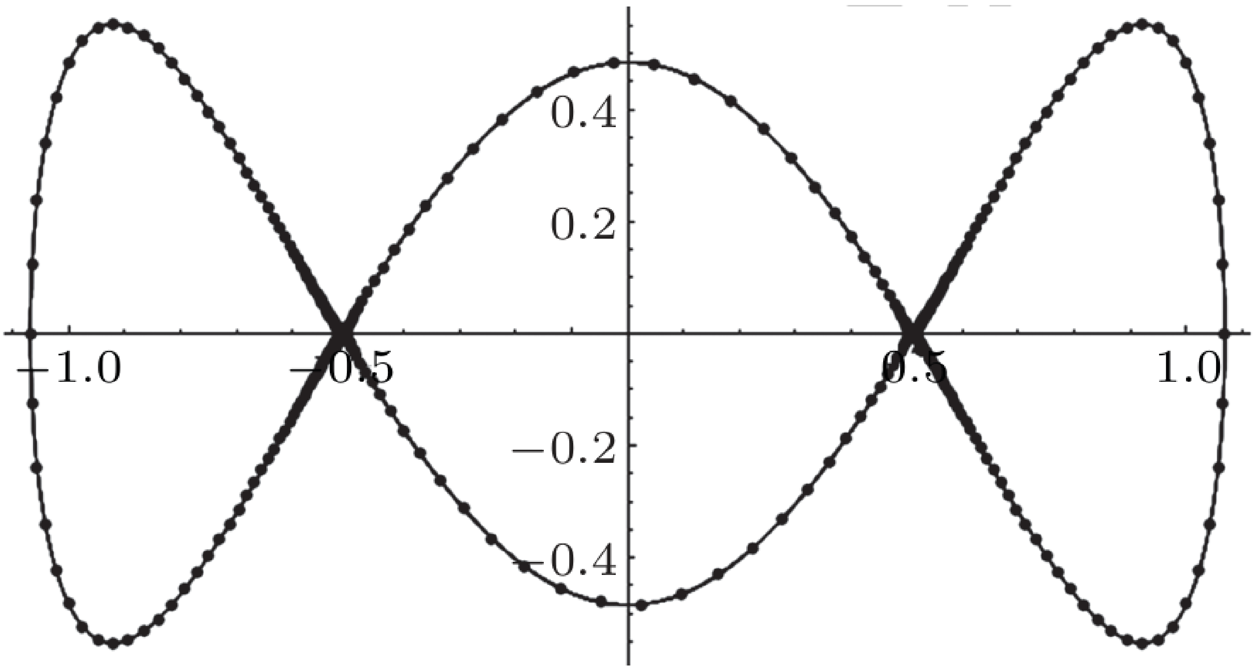

Figure 1: Homoclinic and heteroclinic orbits of the cubic–quintic Duffing oscillator (Original Fig. 1). Solid: Runge–Kutta. Dotted: GPA method.

3.2 Duffing Equation with Φ^6 Potential

$$\ddot{x} + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 = 0. \tag{4}$$

With $c_1 = c_3 = c_5 = 1, c_2 = c_4 = -1.5$, three saddle points allow both homoclinic and heteroclinic orbits. GPA[5/5] and GPA[7/7] outperform the classical Padeé approximant at the same order.

3.3 Generalized Duffing–Harmonic Oscillator

$$\ddot{x} + \frac{\lambda x + \mu x^3}{1 + \nu x^2} = 0. \tag{5}$$

With $\lambda = -1, \mu = 5, \nu = 5$: GPA[3/3] captures two homoclinic orbits. With $\lambda = 3, \mu = -5, \nu = 2$: GPA[8/8] captures heteroclinic orbits.

4 Significance

1. **Unifying framework:** GPA subsumes classical Padeé and all quasi-Padeé variants as special cases.

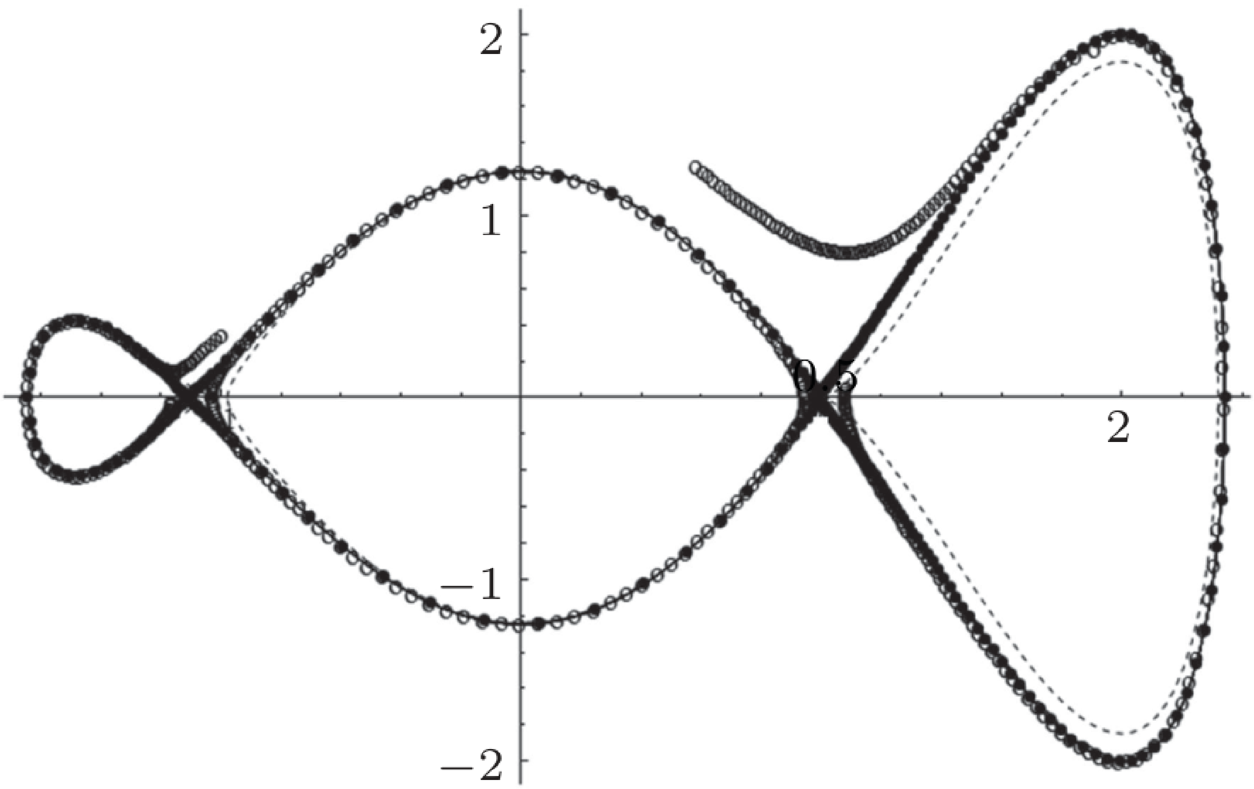

Figure 2: Homoclinic and heteroclinic orbits of the Φ^6 oscillator (Original Fig. 3). Solid: Runge–Kutta. Dotted: GPA. Dashed: classical Padeé approximant.

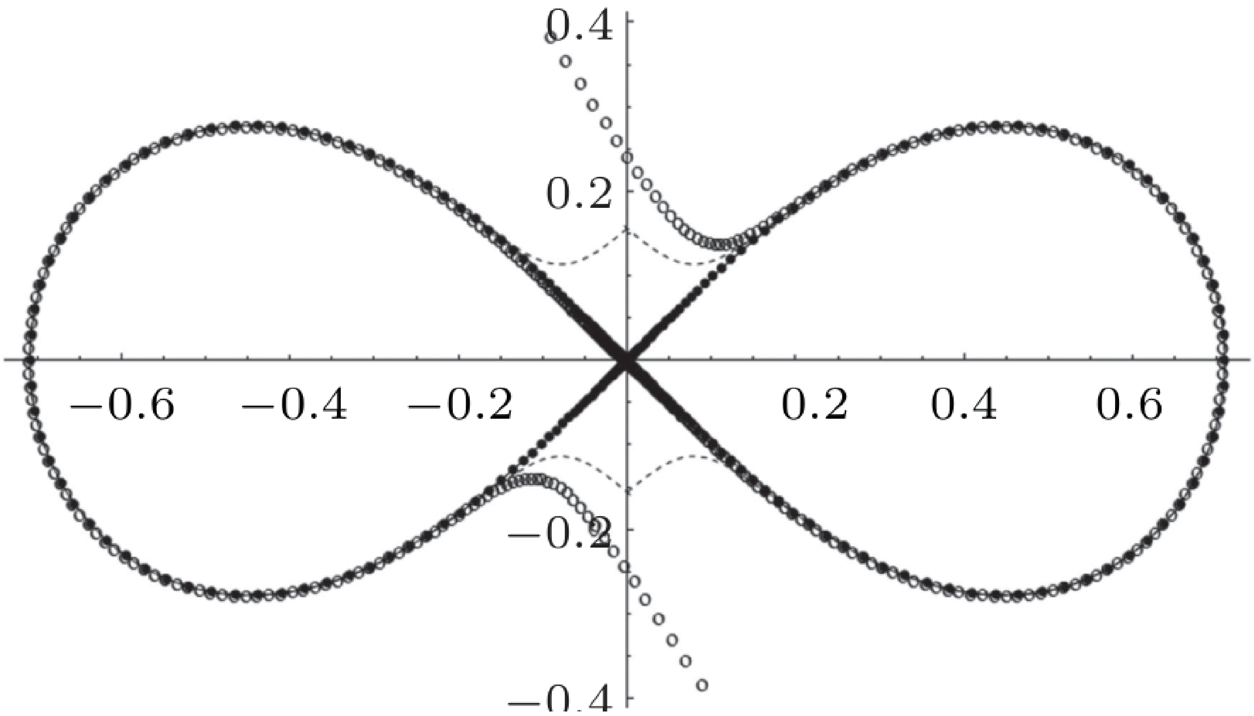

Figure 3: Homoclinic orbits of the Duffing-harmonic oscillator, $\lambda = -1, \mu = 5, \nu = 5$ (Original Fig. 4). Solid: Runge–Kutta. Dotted: GPA[3/3].

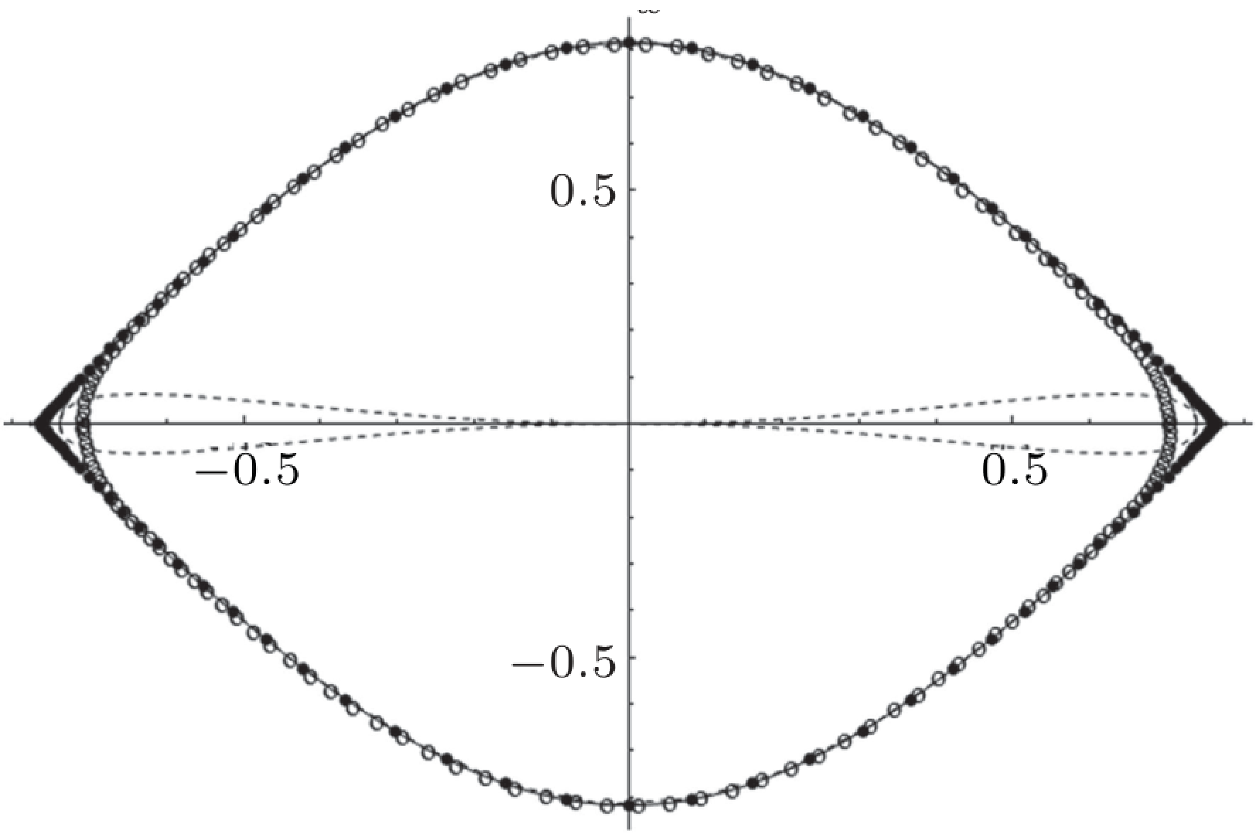

Figure 4: Heteroclinic orbits of the Duffing-harmonic oscillator, $\lambda = 3, \mu = -5, \nu = 2$ (Original Fig. 5). Solid: Runge–Kutta. Dotted: GPA[8/8].

2. **Dual homoclinic–heteroclinic pipeline:** $\operatorname{sech}^i t$ / $\tanh^i t$ dual approximant construction handles both orbit types in one framework.
3. **Superior accuracy:** Hyperbolic-function GPA achieves demonstrably higher accuracy than polynomial-based Padeé at the same order, especially for strongly nonlinear cases.

5 Limitations

- Conservative systems only ($\varepsilon = 0$).
- Approximation order empirical; no systematic error bound.
- High-order coefficient matching may become ill-conditioned.

6 Connection to the GHFP/MGHFP Series

The GPA method and the GHFP/MGHFP family share the mathematical toolkit of rational function approximation but pursue complementary routes: GPA constructs homoclinic/heteroclinic orbits via hyperbolic-function approximants in the time domain; GHFP/MGHFP uses angular-domain perturbation for amplitude–parameter relationships. The $\operatorname{sech}^i t$ basis introduced here also appears in the GHFP homoclinic construction $x = a \sin^2 \varphi + b$.

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